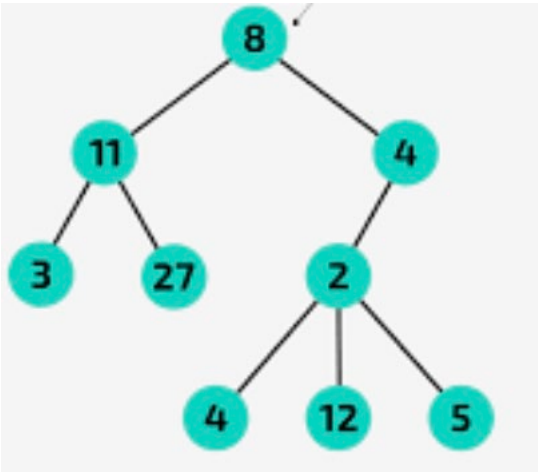


Task #1: Tree traversal

Given:



V: root, L: left, R: right

Pre-order (VLR): _____

IN-order (LVR): _____

Post-order (LRV): _____

 Challenging question:
 Customized traversal for (LVRL):

Given:

Pre-order: C A B E D G F

In-order: B A D E C G F

To draw the tree:

Given:

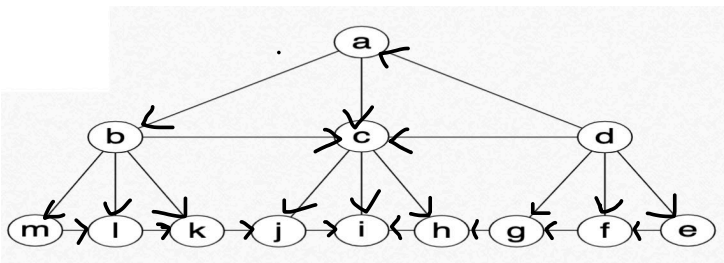
Post-order: D F A B E C

In-order: D F A C E B

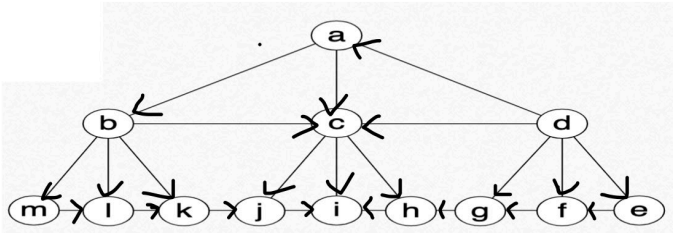
To draw the tree:

Task #2: DFS based Topological Sort

1) Consider this given graph:

DFS order (starting from Vertex d):
_____Order for Vertex become dead-end:
_____Topological Sort order:

2) Consider this given graph:



Default DFS order:

Order for Vertex become dead-end:

Topological Sort order:

3) Consider the same graph as 2)

DFS order (starting from Vertex c):

Order for Vertex gets into the stack:

Topological Sort order:

4) Consider the same graph as 2)

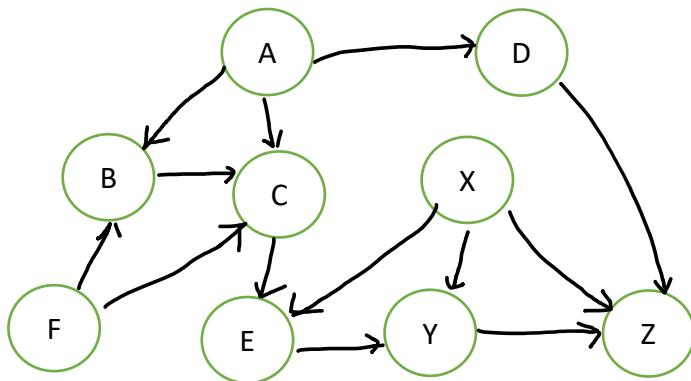
DFS order (starting from Vertex m):

Order for Vertex gets out from the stack:

Topological Sort order:

Task #3: Decrease-by-one Topological Sort

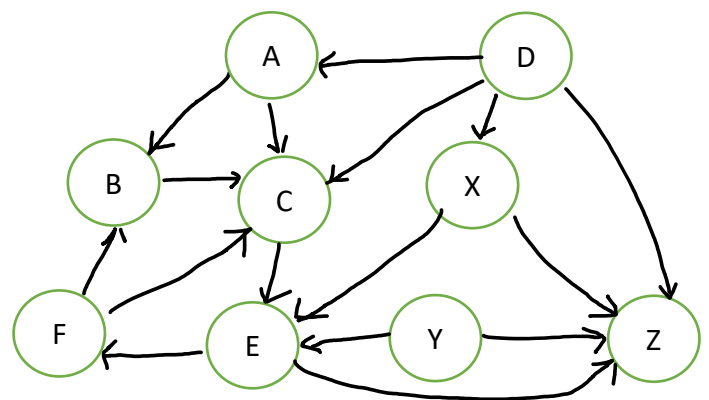
1) Consider this given graph:



Topological Sort order (starting from Vertex F):

Topological Sort order (starting from Vertex X):

2) Consider this given graph:



Topological Sort order (starting from Vertex D):

Topological Sort order (starting from Vertex Y):